MATH1025B
Essential Mathematical Methods

Chapter 4: Binomial Theorem

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Definition 1.1 (Factorials)

The **factorial** of a positive integer n is the number
$n! = n \times (n-1) \times \cdots 2 \times 1 = \prod_{i=1}^{n} i$.
Furthermore, $0!$ is defined to be 1.

A k**-permutation** of n objects is an ordered arrangement of k objects out of n distinct objects.
The number of such k-permutations is denoted variously by the symbols

$$P_k^n,\ {}_nP_k \text{ or } P(n,k).$$

It is computed by the formula

$$P_k^n = n \times (n-1) \times \cdots \times (n-k+1) = \frac{n!}{(n-k)!} \qquad \text{if } 0 \leq k \leq n.$$

Example 1.1

Suppose m girls and n boys are to be seated in a row so that no two boys sit together. If $m > n$, find the number of ways in which they can be seated.

Fix a particular order of the m girls.
Since $m > n$ and no two boys can sit together, there are $(m+1)$ places for n boys as shown in the figure below.

	G_1		G_2		$\cdots$		G_m	

The number of ways for the boys to be seated is P_n^{m+1}.
Now the m girls can be ordered in P_m^m ways, so the number of ways in which the girls and boys can be seated is

$$P_m^m \times P_n^{m+1} = \frac{m!(m+1)!}{(m-n+1)!}.$$

Definition 1.2 (Binomial Coefficients)

Let $n \geq 0$ be an integer. For each integer k, $0 \leq k \leq n$, the **binomial coefficient of n over k** is the number

$$C_k^n := \frac{n!}{k! \cdot (n-k)!}.$$

The number C_k^n is also denoted by ${}_nC_k$, $C(n,k)$ or $\binom{n}{k}$.

Remark:

(i) It turns out that C_k^n is an integer.

(ii) Given a set S of n elements, a **k-combination** of S is a selection of k elements from S such that order does not matter (i.e. it is a k-element subset of S). The number of such k-combinations is

$$\frac{P_k^n}{P_k^k} = \frac{n!}{k!(n-k)!} = C_k^n, \qquad \text{if } 0 \leq k \leq n.$$

Example 1.2

The number of ways to choose 3 letters from A, B, C, D, E (5 letters) in order (without repetition) is $P_3^5 = \dfrac{5!}{(5-3)!} = 60$.

	# ways to choose 3 objects out of 5 regardless of ordering $= C_3^5 = 10$									
# ways to permute 3 objects $= 3! = 6$	ABC	ABD	ABE	ACD	ACE	ADE	BCD	BCE	BDE	CDE
	ACB	ADB	AEB	CAD	CAE	DEA	BDC	BEC	BED	DCE
	BAC	BAD	BAE	DAC	DAE	DAE	CBD	CBE	DBE	DEC
	BCA	BDA	BEA	DCA	EAC	EAD	CDB	CEB	DEB	EDC
	CAB	ABD	EBA	CDA	ECA	AED	DBC	ECB	EBD	CED
	CBA	DBA	EAB	DCA	AEC	DEA	DCB	EBC	EDB	ECD

$$3! \times C_3^5 = P_3^5 = \frac{5!}{(5-3)!}$$

$$\therefore \quad C_3^5 = \frac{5!}{3!(5-3)!}$$

Proposition 1.1

Let $n \geq 0$ be an integer.

(a) $C_0^n = C_n^n = 1$.

(b) *If k is an integer such that $0 \leq k \leq n$, then $C_k^n = C_{n-k}^n$.*

(c) *If k is an integer among $0 \leq k \leq n-1$, then $C_k^n + C_{k+1}^n = C_{k+1}^{n+1}$.*

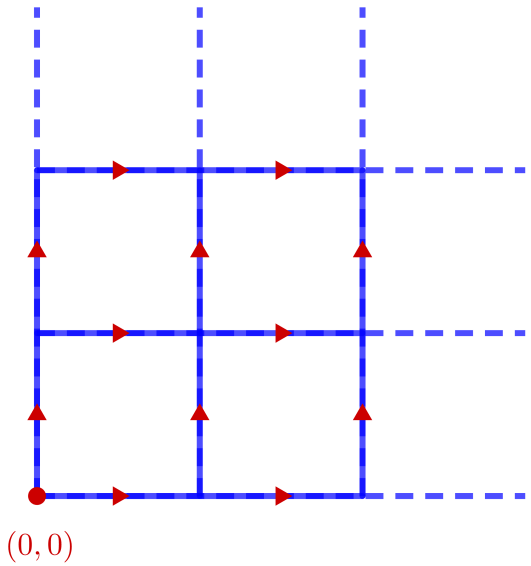
Proof. We only prove (c) here:

$$\begin{aligned} C_k^n + C_{k+1}^n &= \frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} \\ &= \frac{n!}{k!(n-k-1)!}\left(\frac{1}{n-k} + \frac{1}{k+1}\right) \\ &= \frac{n!}{k!(n-k-1)!} \cdot \frac{n+1}{(n-k)(k+1)} \\ &= \frac{(n+1)!}{(k+1)!(n-k)!} = C_{k+1}^{n+1}. \end{aligned}$$

□

Example 1.3

If someone starts from $(0,0)$ and either moves up or to the right by 1 unit every time, how many different ways one can move from $(0,0)$ to $(5,3)$?



He needs to move 8 steps in total, and move to the right 5 times. So there are C_5^8 ways to move to $(5,3)$.
In general, there are C_k^n ways to move to $(k, n-k)$.
Furthermore,

$$\# \text{ ways to } (k+1, n-k) = \# \text{ ways to } (k, n-k) + \# \text{ ways to } (k+1, n-k-1)$$

$$\therefore \quad C_{k+1}^{n+1} = C_k^n + C_{k+1}^n.$$

Pascal's Triangle

$$
\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1 \\
 & & & & \vdots & & & & \\
 & & & & C_0^0 & & & & \\
 & & & C_0^1 & & C_1^1 & & & \\
 & & C_0^2 & & C_1^2 & & C_2^2 & & \\
 & C_0^3 & & C_1^3 & \underbrace{+} & C_2^3 & & C_3^3 & \\
C_0^4 & & C_1^4 & & C_2^4 & & C_3^4 & & C_4^4 \\
 & & & & \vdots & & & &
\end{array}
$$

Binomial Theorem

Question:

How do we expand $(x+y)^n$, where $n \in \mathbb{Z}^+$.

$n = 3$ case:

$$\begin{aligned}(x+y)^3 &= (x+y)(x+y)(x+y)\\ &= (xx+yx+xy+yy)(x+y)\\ &= xxx+yxx+xyx+yyx+xxy+yxy+xyy+yyy\\ &= x^3+3x^2y+3xy^2+y^3.\end{aligned}$$

Observations:

(i) There are $2^3 = 8$ terms in the expansion of $(x+y)^3$ before simplification.

(ii) Every term is product of x's and y's, and exactly one of x or y is taken from each bracket.

(iii) There are three x^2y since we have to pick two x's from two brackets and one y from the remaining bracket and there are $C_2^3 = 3$ ways to do so.

(iv) $$\begin{aligned}(x+y)^3 &= C_3^3x^3y^0 + C_2^3x^2y^1 + C_1^3x^1y^2 + C_0^3x^0y^3 \quad \text{(choosing } x\text{'s)}\\ &= C_0^3x^3y^0 + C_1^3x^2y^1 + C_2^3x^1y^2 + C_3^3x^0y^3 \quad \text{(choosing } y\text{'s).}\end{aligned}$$

(This is expected because $C_k^n = C_{n-k}^n$.)

Theorem 1.2 (Binomial Theorem)

Let n be a positive integer. Then,

$$(x+y)^n = \sum_{k=0}^{n} C_k^n x^k y^{n-k} = C_0^n x^0 y^n + C_1^n x^1 y^{n-1} + \cdots + C_{n-1}^n x^{n-1} y^1 + C_n^n x^n y^0.$$

Proof. Let $P(n)$ be the equality above.

- When $n = 1$, LHS $= (x+y)^1 = x + y = C_0^1 x^0 y^1 + C_1^1 x^1 y^0 =$ RHS. So $P(1)$ is true.
- Assume $P(n)$ is true where $n \geq 1$. Then

$$\begin{aligned}
&(x+y)^{n+1} \\
&= (x+y)^n (x+y) \\
&= \left(C_0^n x^0 y^n + C_1^n x^1 y^{n-1} + \cdots + C_{n-1}^n x^{n-1} y^1 + C_n^n x^n y^0\right)(x+y) \\
&= \begin{array}{lllllllllll} & & + & C_0^n x^1 y^n & + & \cdots & + & C_{n-1}^n x^n y^1 & + & C_n^n x^{n+1} y^0 & + \\ C_0^n x^0 y^{n+1} & & + & C_1^n x^1 y^n & + & \cdots & + & C_n^n x^n y^1 & & & \end{array} \\
&= C_0^{n+1} x^0 y^{n+1} + C_1^{n+1} x^1 y^n + \cdots + C_n^{n+1} x^n y^1 + C_n^{n+1} x^{n+1} y^0.
\end{aligned}$$

  So $P(n+1)$ is true.
- By mathematical induction, $P(n)$ is true for all $n \in \mathbb{Z}^+$. $\square$

Example 1.4

Put $x = y = 1$, we get $\sum_{k=0}^{n} C_k^n = 2^n$. (Compare it with observation (i).)

Put $x = -1, y = 1$, we get $\sum_{k=0}^{n} (-1)^k C_k^n = 0$.

Example 1.5

Find the constant term of $\left(x^2 - \frac{1}{x}\right)^9$.

By the binomial theorem,

$$\left(x^2 - \frac{1}{x}\right)^9 = \sum_{k=0}^{9} C_k^9 (x^2)^k (-\frac{1}{x})^{9-k} = \sum_{k=0}^{9} (-1)^{9-k} C_k^9 x^{3k-9}.$$

We obtain the constant term when $3k - 9 = 0$, that is, $k = 3$.
So the constant term is $(-1)^6 C_3^9 = 84$.

Example 1.6

Let n be a positive integer. By considering $(x+1)^{2n} = (x+1)^n(x+1)^n$, show that

$$C_n^{2n} = C_0^n C_n^n + C_1^n C_{n-1}^n + \cdots + C_n^n C_0^n.$$

By the binomial theorem,

$$(x+1)^{2n} = \sum_{k=0}^{2n} C_k^{2n} x^k.$$

So the coefficient of x^n is C_n^{2n}. On the other hand,

$$(x+1)^n(x+1)^n = \left(\sum_{r=0}^{n} C_r^n x^r\right)\left(\sum_{k=0}^{n} C_k^n x^k\right) = \sum_{r=0}^{n}\sum_{k=0}^{n} \underbrace{C_r^n C_k^n x^{r+k}}_{x^n \text{ term when } r+k=n}.$$

So the coefficient of x^n is $C_0^n C_n^n + C_1^n C_{n-1}^n + \cdots + C_n^n C_0^n$.
Therefore, $C_n^{2n} = C_0^n C_n^n + C_1^n C_{n-1}^n + \cdots + C_n^n C_0^n$.

What about C_m^{2n} for m other than n?

Exercise 1.1 (Vandermonde's identity)

For any nonnegative integers m, n and r such that $m, n \geq r$, show that

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k} \binom{n}{r-k}.$$

Example 1.7

Let $n \in \mathbb{Z}^+$ and let y be a real constant. Then

$$(x+y)^n = \sum_{k=0}^{n} C_k^n x^k y^{n-k} \qquad \text{for any } x \in \mathbb{R}.$$

Differentiate both sides with respect to x:

$$\frac{d}{dx}(x+y)^n = \frac{d}{dx}\left(\sum_{k=0}^{n} C_k^n x^k y^{n-k}\right) = \sum_{k=0}^{n} \frac{d}{dx}\left(C_k^n x^k y^{n-k}\right)$$

$$\begin{aligned} n(x+y)^{n-1} &= \sum_{k=0}^{n} kC_k^n x^{k-1} y^{n-k} \\ &= \sum_{k=1}^{n} kC_k^n x^{k-1} y^{n-k}. \qquad (kC_k^n x^{k-1} y^{n-k} = 0 \text{ when } k=0) \end{aligned}$$

Put $x = y = 1$, we get $\displaystyle\sum_{k=1}^{n} kC_k^n = n2^{n-1}$.

Exercise 1.2

Let $n \in \mathbb{Z}^+$. Show that $\displaystyle\sum_{k=0}^{n} \frac{(-1)^k}{k+1} C_k^n = \frac{1}{n+1}$.

Exercise 1.3

(a) Let $n, k \in \mathbb{Z}^+$ such that $1 \leq k \leq n$. Show that $k \cdot C_k^n = n \cdot C_{k-1}^{n-1}$.

(b) Let $n \in \mathbb{Z}^+$. Hence, or otherwise, find the values of the following sums.

(i) $\displaystyle\sum_{k=0}^{n} kC_k^n$

(ii) $\displaystyle\sum_{k=0}^{n} k(-1)^k C_k^n$

(iii) $\displaystyle\sum_{k=0}^{n} \frac{1}{k+1} C_k^n$

(iv) $\displaystyle\sum_{k=0}^{n} k^2 C_k^n$

Ans: (i) $n \cdot 2^{n-1}$, (ii) 0 if $n \geq 2$; -1 if $n = 1$, (iii) $\frac{1}{n+1}(2^{n+1} - 1)$, (iv) $n(n+1) \cdot 2^{n-2}$.

Example 1.8 (Binomial Distribution)

Let $0 < p < 1$ be the probability of getting a head when a biased coin is tossed. Suppose the coin is tossed n times. Let X be the random variable that denotes the number of heads obtained. Then $X \sim B(n, p)$, that is,

$$P(X = k) = C_k^n p^k (1-p)^{n-k}, \qquad 0 \le k \le n.$$

$$\text{Expected number of heads} = \sum_{k=0}^{n} kP(X = k) = \sum_{k=0}^{n} kC_k^n p^k (1-p)^{n-k}.$$

From Example 1.7, we have

$$n(x+y)^{n-1} \cdot x = \left(\sum_{k=0}^{n} kC_k^n x^{k-1} y^{n-k} \right) x = \sum_{k=0}^{n} kC_k^n x^k y^{n-k}.$$

Put $x = p$ and $y = 1 - p$, we have $\displaystyle\sum_{k=0}^{n} kC_k^n p^k (1-p)^{n-k} = np$.

Exercise 1.4

Show that the variance $\sum_{k=0}^{n}(k-np)^2 P(X=k) = np(1-p)$.

Example 1.9

If $X \sim B(n,p)$, then as k goes from 0 to n, $P(X=k)$ first increases monotonically and then decreases monotonically, reaching its largest value when k is the largest integer less than or equal to $(n+1)p$.

$$
\begin{aligned}
& & P(X=k-1) &\leq P(X=k) \\
&\iff & C_{k-1}^{n} p^{k-1}(1-p)^{n-k+1} &\leq C_{k}^{n} p^{k}(1-p)^{n-k} \\
&\iff & \frac{n!}{(k-1)!(n-k+1)!}(1-p) &\leq \frac{n!}{k!(n-k)!}p \\
&\iff & k(1-p) &\leq (n-k+1)p \\
&\iff & k &\leq (n+1)p
\end{aligned}
$$

Example 1.10

Let $a > 0$. Show that there is a constant $C > 0$ such that $n^2(1+a)^{-n} \leq C$ for all integers $n \geq 2$.

Want: find a constant $C' > 0$ such that

$$(1+a)^n \geq C' n^2 \qquad \forall\, n \geq 2.$$

By the binomial theorem, for $n \geq 2$,

$$\begin{aligned}(1+a)^n &= \sum_{k=0}^{n} C_k^n a^k = 1 + C_1^n a + C_2^n a^2 + \cdots + C_n^n a^n \\ &\geq C_2^n a^2 = \frac{n(n-1)}{2} a^2. \qquad (\because C_k^n a^k \geq 0)\end{aligned}$$

For $n \geq 2$, we have

$$n^2 \geq 2n \implies 2n^2 - 2n \geq n^2 \implies n(n-1) \geq \frac{n^2}{2}.$$

Combining the inequalities, we have

$$(1+a)^n \geq \frac{a^2}{4} n^2 \qquad \forall\, n \geq 2.$$

Can you do the same for $n^3(1+a)^{-n}$?

Exercise 1.5

(a) Show that for $k, r, n \in \mathbb{Z}^+$ such that $1 \leq k \leq r \leq n$, $C_r^n C_k^r = C_k^n C_{r-k}^{n-k}$.

(b) Hence, show that (i) $\displaystyle\sum_{r=k}^{n} C_r^n C_k^r = 2^{n-k} C_k^n$, (ii)

$$\sum_{r=k}^{n} (-1)^r C_n^r C_k^r = \begin{cases} 0 & \text{if } 1 \leq k < n \\ (-1)^n & \text{if } k = n \end{cases}.$$

Exercise 1.6

Let $n \in \mathbb{Z}^+$.

(a) Show by Cauchy Schwarz inequality that
$\sqrt{n(2^n - 1)} > \sqrt{C_1^n} + \sqrt{C_2^n} + \cdots + \sqrt{C_n^n}$.

(b) By considering $n(x+1)^{2n-1} = n(x+1)^{n-1}(x+1)^n$, find $\sum_{r=1}^{n} r\,(C_r^n)^2$.

Multinomial Theorem

Question:

In general, what is the expansion of $(x_1 + x_2 + \cdots + x_m)^n$ where $m, n \in \mathbb{Z}^+$?

Theorem 1.3 (Multinomial Theorem)

Let $m, n \in \mathbb{Z}^+$. Then

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{\substack{k_1+k_2+\cdots+k_m=n \\ k_1,k_2,\ldots,k_m \geq 0}} \binom{n}{k_1, k_2, \ldots, k_m} x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m},$$

where $\binom{n}{k_1, k_2, \ldots, k_m} := \dfrac{n!}{k_1! k_2! \cdots k_m!}$ *is a multinomial coefficient.*

Remark: $\binom{n}{k_1, k_2, \ldots, k_m}$ is the number of ways to partition n distinct objects into m distinct groups such that the i-th group contains k_i objects, $1 \leq i \leq m$.

Example 1.11

Given n objects and three boxes, labelled A, B and C. Suppose we are to place a given number a of the objects in box A, a given number b in box B, and the remaining $c = n - a - b$ in box C. Assuming the order of placement of objects in each box is irrelevant, what is the number of such placement?

Each placement can be described by first selecting a objects from the set of n for box A, and then picking b objects from the remaining $n - a$ for box B. The number of such placement is

$$\binom{n}{a} \cdot \binom{n-a}{b} \cdot \binom{n-a-b}{c} = \frac{n!}{a!(n-a)!} \cdot \frac{(n-a)!}{b!(n-a-b)!} \cdot 1$$

$$= \frac{n!}{a!\,b!\,c!} = \binom{n}{a, b, c}.$$

Example 1.12

How many ways can you arrange the letters in "$GOOGOL$"?

First arrange the 6 letters as if they were distinct:

$$G_1O_1O_2G_2O_3L$$

The number of ordered arrangement is $6!$.

The number of permutations within O_1, O_2, O_3 is $3!$.
The number of permutations within G_1, G_2 is $2!$.
The number of permutations within L is $1!$.

Hence, the number of ways we can arrange the letters in "$GOOGOL$" is

$$\frac{6!}{3!\,2!\,1!} = \binom{6}{3,2,1} = 60.$$